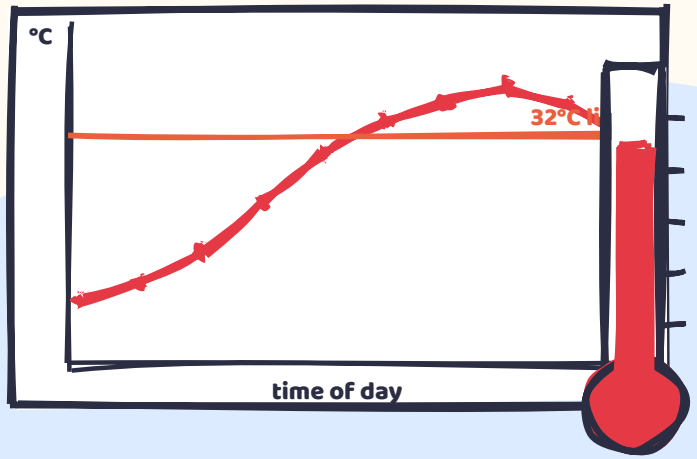
Young Innovators · Class 21

Python Data Lab: Is Our Classroom Too Hot?

Real data. Real question. Real answer.

Heat + humidity | CSV + stats | Charts



Did you know?
Studies show students score lower in tests when classrooms get too hot. Heat is a learning problem, not just a comfort problem.

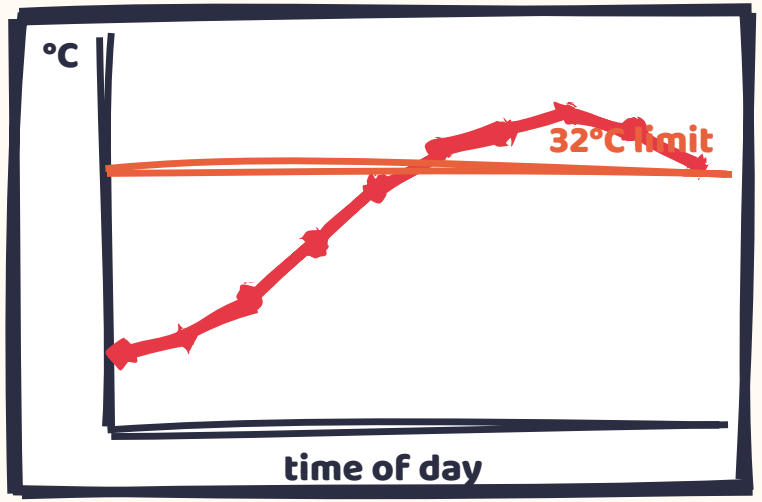
LEARN

Let's think...

Is it really too hot, or does it just feel like it?

The great fan debate

Every afternoon someone says 'It's SO hot in here!' and someone else says 'It's fine.' Who is right? Opinions argue. **Data** decides. Today we let the numbers settle it.



It's fine!

Way too hot!

Show me the data!

Today: you analyse our **classroom temperature and humidity log** in Python and present a chart with a conclusion.

Learning objectives

By the end of this class I can...

1	**Comfort science:** explain why **temperature + humidity** together decide how hot we feel	**Proof:** I can say why 34 °C in Karachi feels worse than 34 °C in Quetta
2	**Read the data:** load a **CSV** file in Python and calculate the **average, max and min**	**Proof:** my program prints correct stats for our classroom
3	**Chart + conclusion:** plot a labelled **matplotlib** chart and write a conclusion backed by **evidence**	**Proof:** my chart answers the question in one sentence

LEARN

Objective 1 · Comfort science · video

Video time!



Why Does Humidity Make It Feel Hotter?

SciShow · about 3 minutes · youtu.be/SGHRz8wpj3E

Watch for...

1. Why does humid air make us feel **hotter**?
2. What does **sweat** do to cool us down?
3. What is the **heat index**?

Scan to watch

After the video: Is Karachi or Quetta harder to cool down in on a 34 °C day? Why?

Objective 1 · Comfort science

Humidity changes how hot we feel

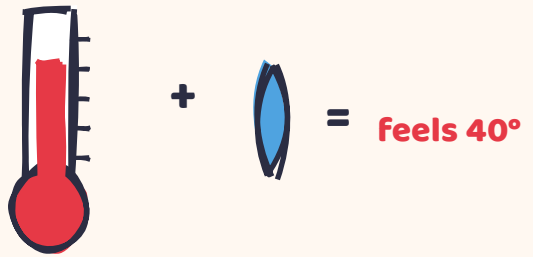
The idea

Our body cools by **sweating**: sweat evaporates and takes heat away. In **humid** air, sweat evaporates slowly, so we stay hot.

The **heat index** combines both: 32 °C at 40% humidity feels about 33 °C; at 70% it feels about **40 °C**!

Comfortable classroom: about **20 to 26 °C**, humidity **40 to 60%**.

Example: wet clothes on a line



Clothes dry fast on a dry, windy Quetta day and slowly on a sticky Karachi day. Sweat on your skin is the same: humid air = slow drying = hot body.

Objective 2 · Read the data

Data lives in a CSV file

The idea

Our DHT22 logger saved a reading every 30 minutes as a **CSV** (comma-separated values) file. Each line is one **row**; commas split the **columns**:

```
time,temp,humidity
08:00,27.0,62
08:30,27.5,60
```

Python's **csv** module reads it row by row. Numbers arrive as **text**, so we convert with **float()**.

Example: a school register

```
time,temp,hum
08:00,27.0,62
08:30,27.5,60
09:00,28.5,58
...
```

The class register is a table: one row per student, one column per day. A CSV is the same table, typed as plain text with commas.

LEARN

Objective 3 · Chart + conclusion

A good chart answers one question

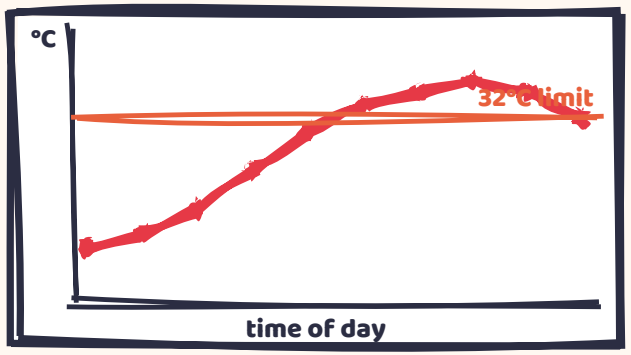
The idea

Line chart: how something changes over **time**. **Bar chart**: compare **categories** (Room A vs Room B).

Every chart needs: a **title** that is the question, **axis labels with units**, and a **reference line** (e.g. 26 °C comfort limit) so the reader sees the answer instantly.

Then one sentence: **claim + evidence**.

Example: a weather forecast



TV weather shows a line of temperatures through the day with sunny icons. Nobody reads a table of numbers; the chart tells the story in 2 seconds.

Real world

Data that decides



Heatwave warnings
PMD warns cities when the heat index is dangerous



Hospitals
Plan for more patients during heatwaves



Farmers
Choose when to water using temperature data



Climate science
100 years of readings show our planet warming



If our data shows the room is too hot after 12:00, what could the school change?

Evidence wins
A chart with real data is how you persuade a principal, a city, or a government.

Quick quiz

Data detective

Question	Answer
1. Readings: 28, 30, 32. Average?	
2. 34 °C, humidity 70%: feels hotter or cooler than 34 °C?	
3. Line or bar chart for temperature over a day?	
4. Why float(row[1]), not row[1]?	
5. What must every axis label include?	

Answers

Cover! 1: 30 · 2: hotter · 3: line · 4: CSV values are text · 5: the unit (°C, %, time)

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human line graph
2. Our data lab
3. Load the CSV and get the stats
4. Plot it with matplotlib
5. Write the conclusion
6. Check your analysis
7. Data scientist challenges

MAKE

Class game

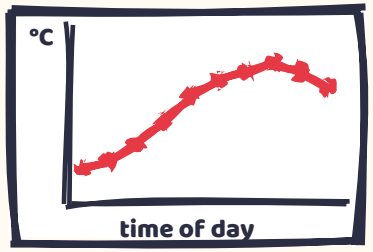
Human line graph

How to play

1. Teacher reads 8 temperatures from the log, one per time.
2. 8 students line up left to right (time). Each stands on a step marked on the floor for their temperature.
3. Hold a rope from student to student: you are the line chart!
4. Where is the peak? Where does it cross the 26 °C tape?

Then discuss

What happened between 11:00 and 13:00? What might cause it?



Big idea

A line chart is just points joined in time order.

Build it

Our data lab

Thonny
Python editor

matplotlib
Tools → Manage packages → install

classroom_log.csv
From our DHT22 logger (class folder)

Room B log (optional)
To compare two rooms

Worksheet
For your conclusion

Projector
For the data showcase

Put the CSV in the SAME folder as your .py file, or Python can't find it.

Bad data
Real data has mistakes: a sensor glitch might log 85 °C. Spot it and decide what to do before you calculate.

MAKE

Objective 2 · Read the data · Build step 1 of 3

Load the CSV and get the stats

```
import csv

times, temps, hums = [], [], []
with open("classroom_log.csv") as f:
    reader = csv.reader(f)
    next(reader)                        # skip the header row
    for row in reader:
        times.append(row[0])
        temps.append(float(row[1]))
        hums.append(float(row[2]))

avg = sum(temps) / len(temps)
print("Readings:", len(temps))
print("Average:", round(avg, 1), "°C")
print("Max:", max(temps), "°C at", times[temps.index(max(temps))])
print("Min:", min(temps), "°C")

hot = [t for t in temps if t > 26]
print("Too hot:", len(hot), "of", len(temps), "readings",
      f"({len(hot) / len(temps):.0%})")
```

Lists

Each column becomes a list. temps[0] is the first reading.

index()

temps.index(max(temps)) finds WHERE the max is, so we can look up its time.

Check

Do your numbers make sense? Average between min and max?

MAKE

Objective 3 · Chart + conclusion · Build step 2 of 3

Plot it with matplotlib

```
import matplotlib.pyplot as plt

fig, ax1 = plt.subplots(figsize=(10, 5))
ax1.plot(times, temps, "o-", color="tab:red", label="Temperature")
ax1.axhline(26, linestyle="--", color="orange", label="Comfort limit 26 °C")
ax1.set_xlabel("Time of day")
ax1.set_ylabel("Temperature (°C)")
ax1.set_title("Is our classroom too hot? (Mon 14 Oct)")

ax2 = ax1.twinx()                        # second y-axis
ax2.plot(times, hums, "s:", color="tab:blue", label="Humidity")
ax2.set_ylabel("Humidity (%)")

fig.legend(loc="upper left")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("classroom_chart.png", dpi=150)
plt.show()
```

Reference line

axhline() draws the comfort limit, so the answer jumps out.

twinx()

Two y-axes on one chart: °C on the left, % on the right.

Save

savefig() makes a PNG for your report.

MAKE

Objective 3 · Build step 3 of 3

Write the conclusion

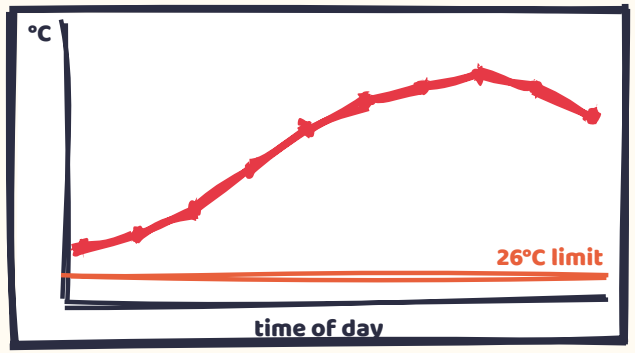
a Look at your chart. Answer the question in **one sentence**: yes / no / sometimes.

b Add **evidence**: 'The room was above 26 °C for 60% of readings, peaking at 33.5 °C at 13:30.'

c Add a **reason**: what might cause it? (sun on windows, 35 people, fans off…)

d Add a **recommendation** the school can act on.

e Add one **limitation**: only one day? one sensor? near a window?



Check: Your report has claim, evidence, reason, recommendation and limitation.

Test like an engineer

Check your analysis

Check	Done?	Notes
Stats printed (avg, max, min)		
Glitch values found and handled		
Chart has title, labels, units		
Reference line shown		
Conclusion with evidence		

Debug checklist

FileNotFoundError? CSV not in the same folder.
ValueError: could not convert? A blank or 'nan' row: skip it with try/except.
Squashed x-labels? plt.xticks(rotation=45).

Ask a partner to read only your chart for 5 seconds. Can they tell you the answer?

MAKE

Level up!

Data scientist challenges

Choose your challenge

Heat index: calculate 'feels like' for each reading with a heat index formula and plot it.

Two rooms: plot Room A and Room B on one chart. Which is worse, and why?

Live data: log new readings from the ESP32 straight to a CSV and update the chart.

Bar chart: average temperature per hour as a bar chart.

Think like an engineer

Can one day of data prove the room is always too hot? What would make your evidence stronger?

Boss level

Send your chart and conclusion to the principal as a one-page report.

Objective check

Can I really do it? Prove it.

01	Why can 32 °C feel like 40 °C? Explain with sweat.	○ ○ ○
02	Run your program. What are the average, max and min, and when was the max?	○ ○ ○
03	Show your chart and read your conclusion with evidence.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Data showcase: each group presents its chart in 60 seconds. The class votes for the most convincing evidence.

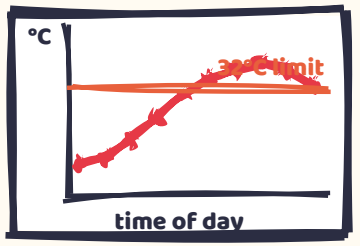
Show what you know

What did we learn?

- [x] Humidity stops sweat evaporating, so we feel hotter
- [x] CSV = rows and columns of plain text
- [x] float() turns text into numbers
- [x] Line charts show change over time
- [x] Claim + evidence + limitation = a strong conclusion

Exit ticket: what does a reference line on a chart help the reader see?

At home
Check the weather app at home: what is the 'feels like' temperature today, and why is it different?



Next class:
Smart Posture Corrector Wearable!